DSBDA Viva Questions and Answers (SPPU 2019 Pattern)

This PDF contains viva questions and answers based on DSBDA practical codes including Pandas, NumPy, Visualization, Machine Learning, NLP, and Seaborn.

Q1. What is Pandas?

Answer: Pandas is a Python library used for data analysis and handling tabular data.

Q2. What is a DataFrame?

Answer: A DataFrame is a 2D table-like structure in Pandas with rows and columns.

Q3. What is NumPy?

Answer: NumPy is a Python library used for numerical and mathematical operations.

Q4. What is Matplotlib?

Answer: Matplotlib is a Python library used for creating graphs and charts.

Q5. What is Seaborn?

Answer: Seaborn is a statistical visualization library built on top of Matplotlib.

Q6. What does pd.read_csv() do?

Answer: It reads a CSV file and loads it into a DataFrame.

Q7. What does df.head() do?

Answer: It displays the first 5 rows of the dataset.

Q8. What does df.tail() do?

Answer: It displays the last 5 rows of the dataset.

Q9. What is df.shape?

Answer: It returns the number of rows and columns in the dataset.

Q10. What is df.dtypes?

Answer: It shows the datatype of each column.

Q11. What is df.info()?

Answer: It gives summary information about the DataFrame.

Q12. What is df.describe()?

Answer: It gives statistical summary of numeric columns.

Q13. What is the use of isnull()?

Answer: It checks for missing values in the dataset.

Q14. What does sum() do after isnull()?

Answer: It counts total missing values column-wise.

Q15. What is dropna()?

Answer: It removes rows containing missing values.

Q16. What is fillna()?

Answer: It fills missing values with specified values.

Q17. What is mean?

Answer: Mean is the average value of data.

Q18. What is median?

Answer: Median is the middle value after sorting data.

Q19. What is mode?

Answer: Mode is the most frequently occurring value.

Q20. What is standard deviation?

Answer: It measures spread or variability in data.

Q21. What is variance?

Answer: Variance measures how far values are spread from mean.

Q22. What is outlier?

Answer: An outlier is an abnormal or extreme value in data.

Q23. What is IQR?

Answer: IQR is Interquartile Range used to detect outliers.

Q24. Formula of IQR?

Answer: IQR = Q3 - Q1.

Q25. What is Q1?

Answer: Q1 is the 25th percentile.

Q26. What is Q3?

Answer: Q3 is the 75th percentile.

Q27. Why detect outliers?

Answer: To improve data quality and model accuracy.

Q28. What is a histogram?

Answer: Histogram shows frequency distribution of data.

Q29. What is a boxplot?

Answer: Boxplot shows median, quartiles, and outliers.

Q30. What is a scatter plot?

Answer: Scatter plot shows relationship between two variables.

Q31. What is pairplot?

Answer: Pairplot shows scatter plots between all features.

Q32. What is heatmap?

Answer: Heatmap is a color representation of data or correlation.

Q33. What is correlation?

Answer: Correlation measures relationship between variables.

Q34. Range of correlation?

Answer: Correlation ranges from -1 to +1.

Q35. What is sns.heatmap()?

Answer: It creates heatmap visualization using Seaborn.

Q36. What is sns.distplot()?

Answer: It plots distribution of continuous data.

Q37. What is jointplot?

Answer: Jointplot combines scatter plot and histogram.

Q38. What is rugplot?

Answer: Rugplot displays individual data points on an axis.

Q39. What is violinplot?

Answer: Violinplot shows distribution and density of data.

Q40. What is stripplot?

Answer: Stripplot shows individual observations.

Q41. What is countplot?

Answer: Countplot shows frequency count of categories.

Q42. What is hue in Seaborn?

Answer: Hue separates data based on categories using colors.

Q43. What is jitter in stripplot?

Answer: Jitter spreads points to avoid overlapping.

Q44. What is groupby()?

Answer: groupby() groups rows based on categories.

Q45. What is cumulative sum?

Answer: Cumulative sum gives running total of values.

Q46. What is value_counts()?

Answer: It counts unique values in a column.

Q47. What is train_test_split()?

Answer: It splits dataset into training and testing data.

Q48. Why split dataset?

Answer: To train and evaluate machine learning models.

Q49. What is Linear Regression?

Answer: Linear Regression predicts continuous numeric values.

Q50. What is Logistic Regression?

Answer: Logistic Regression is used for classification problems.

Q51. What is Naive Bayes?

Answer: Naive Bayes is a probabilistic classification algorithm.

Q52. Difference between GaussianNB and MultinomialNB?

Answer: GaussianNB handles continuous data while MultinomialNB handles discrete counts.

Q53. What is fit()?

Answer: fit() trains the machine learning model.

Q54. What is predict()?

Answer: predict() gives predicted output from trained model.

Q55. What is regression.coef_?

Answer: It returns coefficients of independent variables.

Q56. What is regression.intercept_?

Answer: It returns intercept value of regression line.

Q57. What is R2 score?

Answer: R2 score measures accuracy of regression model.

Q58. Range of R2 score?

Answer: R2 score ranges from 0 to 1.

Q59. What is MAE?

Answer: MAE is Mean Absolute Error.

Q60. What is MSE?

Answer: MSE is Mean Squared Error.

Q61. What is RMSE?

Answer: RMSE is Root Mean Squared Error.

Q62. What is confusion matrix?

Answer: It shows actual vs predicted classifications.

Q63. What is classification report?

Answer: It shows precision, recall, f1-score, and accuracy.

Q64. What is precision?

Answer: Precision measures correctness of positive predictions.

Q65. What is recall?

Answer: Recall measures how many actual positives were identified.

Q66. What is F1-score?

Answer: F1-score is harmonic mean of precision and recall.

Q67. What is NLP?

Answer: NLP stands for Natural Language Processing.

Q68. What is tokenization?

Answer: Tokenization splits text into words or sentences.

Q69. What is stemming?

Answer: Stemming reduces words to root form.

Q70. Example of stemming?

Answer: Cleaning becomes clean.

Q71. What is lemmatization?

Answer: Lemmatization converts word into meaningful root word.

Q72. Difference between stemming and lemmatization?

Answer: Stemming may produce non-meaningful roots while lemmatization gives meaningful roots.

Q73. What are stopwords?

Answer: Stopwords are common words removed during text preprocessing.

Q74. What is POS tagging?

Answer: POS tagging identifies grammatical category of words.

Q75. What is TF-IDF?

Answer: TF-IDF measures importance of words in documents.

Q76. What is corpus?

Answer: Corpus is a collection of text documents.

Q77. What is word_tokenize()?

Answer: It splits sentence into words.

Q78. What is sent_tokenize()?

Answer: It splits paragraph into sentences.

Q79. What is PorterStemmer?

Answer: It is a stemming algorithm in NLTK.

Q80. What is WordNetLemmatizer?

Answer: It performs lemmatization using WordNet.

Q81. What is FacetGrid?

Answer: FacetGrid creates multiple plots for categorical visualization.

Q82. What is dataset preprocessing?

Answer: It is cleaning and transforming raw data before analysis.

Q83. Why normalize data?

Answer: Normalization scales data into similar range.

Q84. What is feature selection?

Answer: Selecting important columns for machine learning.

Q85. What are independent variables?

Answer: Input variables used for prediction.

Q86. What is dependent variable?

Answer: Output variable predicted by model.

Q87. What is supervised learning?

Answer: Learning using labeled data.

Q88. What is classification?

Answer: Classification predicts categorical outputs.

Q89. What is regression?

Answer: Regression predicts continuous outputs.

Q90. What is Iris dataset?

Answer: It is a famous flower dataset used for ML practice.

Q91. What is Titanic dataset?

Answer: Titanic dataset contains passenger survival information.

Q92. What is Social_Network_Ads dataset?

Answer: It contains age, salary, and purchase details.

Q93. What is HousingData dataset?

Answer: It contains house-related features and prices.

Q94. What is Academic Performance dataset?

Answer: Dataset containing student marks and attendance.

Q95. Why use visualization?

Answer: Visualization helps understand patterns and trends in data.